



**TEMASEK JUNIOR COLLEGE**  
**Preliminary Examinations 2009**



**BIOLOGY**  
**Higher 2**  
**Paper 1 Multiple Choice**

**9747/01**

**Friday, 2 October**

**1 hour 15 minutes**

**Additional Materials: Multiple Choice Answer Sheet**

**READ THESE INSTRUCTIONS FIRST**

Write in a soft pencil

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet in the spaces provided.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in soft pencil on the separate Answer Sheet

**Read the instructions on the Answer Sheet very carefully.**

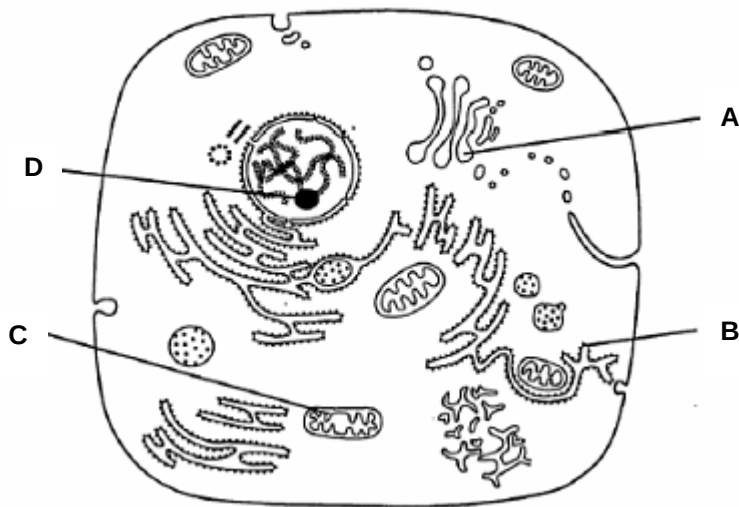
Each correct answer will score one mark. A mark will not be deducted for the wrong answer.  
Any rough working should be done in this booklet.

This document consists of **16** printed pages.

**[Turn over**

1. Which group of structures would be clearly seen in a suitably stained plant cell under the light microscope?
- A. Chloroplasts, mitochondria, nuclei and ribosomes
  - B. Chloroplasts, nuclei, starch grains and vacuole**
  - C. Endoplasmic reticulum, Golgi apparatus, mitochondria and nuclei
  - D. Endoplasmic reticulum, ribosomes, starch grains and vacuole

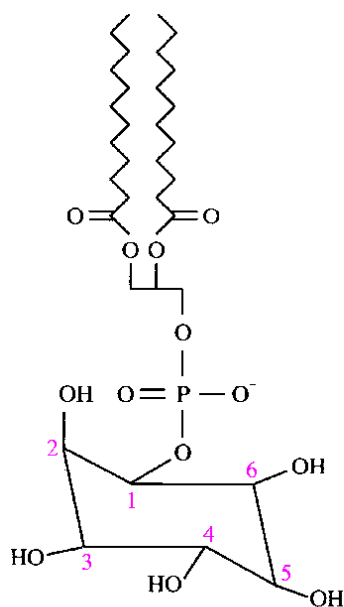
Refer to the diagram shown in below for both questions 2 and 3.



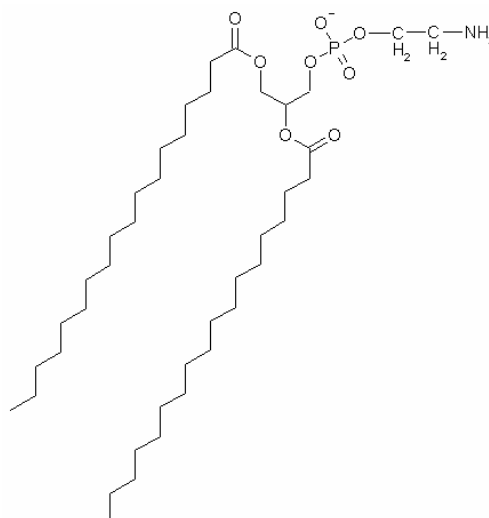
2. Which of the structures, **A**, **B**, **C** or **D** does **not** contain nucleic acid?
3. Which of the following combination of statements is true of structure **C**?

|           | It is the site of carbon dioxide production | It is the site of glycolysis | It is surrounded by a double unit membrane |
|-----------|---------------------------------------------|------------------------------|--------------------------------------------|
| <b>A.</b> | No                                          | No                           | Yes                                        |
| <b>B.</b> | No                                          | Yes                          | Yes                                        |
| <b>C.</b> | <b>Yes</b>                                  | <b>No</b>                    | <b>Yes</b>                                 |
| <b>D.</b> | Yes                                         | Yes                          | Yes                                        |

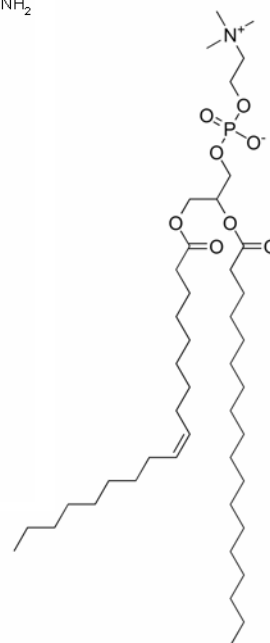
4. The diagram below shows three different phospholipids: phosphatidylcholine, phosphatidylethanolamine and phosphatidylinositol.



**phosphatidylinositol**



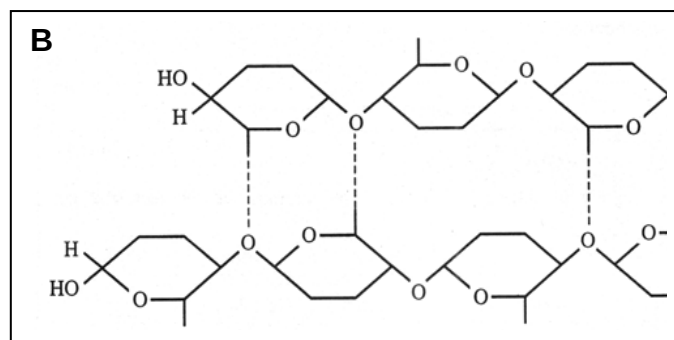
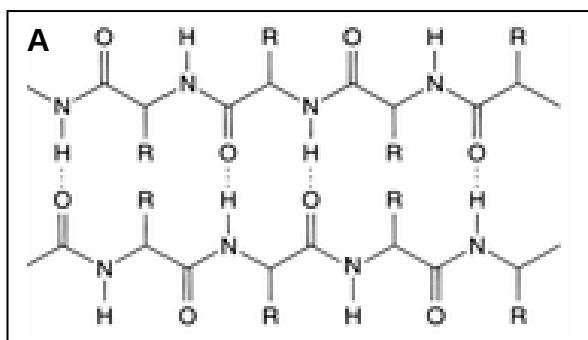
**phosphatidylethanolamine**



**phosphatidylcholine**

Which of the following statements is correct?

- A. **Membranes with a higher phosphatidylcholine content are more fluid.**
  - B. Phosphatidylinositol is unable to form a bilayer in an aqueous environment.
  - C. Only phosphatidylethanolamine shows ambivalent behaviour towards water.
  - D. Only phosphatidylinositol has two fatty acid side chains.
5. The following shows two molecules that perform structural functions.



Which of the following statements does **not** describe a similarity between molecules **A** and **B**?

- A. Hydrogen bonds form cross-linkages between neighboring chains.
- B. **Monomers are joined together via peptide bonds.**
- C. Monomers can be extracted by hydrolysing the bonds within each chain.
- D. Covalent bonds are present in both molecules.

6. Which of the following options correctly matches the functional and structural features of cellulose, collagen, glycogen and triglycerides?

|           |              | Function      | Structure |                                          |                 |
|-----------|--------------|---------------|-----------|------------------------------------------|-----------------|
|           |              |               | Fibrous   | Molecule held together by hydrogen bonds | Branched chains |
| <b>A.</b> | Cellulose    | Support       | ✓         | ✗                                        | ✓               |
|           | Collagen     | Strengthening | ✓         | ✓                                        | ✗               |
| <b>B.</b> | Cellulose    | Support       | ✓         | ✓                                        | ✗               |
|           | Triglyceride | Energy source | ✗         | ✗                                        | ✗               |
| <b>C.</b> | Collagen     | Strengthening | ✓         | ✓                                        | ✓               |
|           | Glycogen     | Storage       | ✗         | ✗                                        | ✓               |
| <b>D.</b> | Glycogen     | Storage       | ✗         | ✓                                        | ✓               |
|           | Triglyceride | Energy source | ✗         | ✓                                        | ✗               |

7. The following table shows some differences between glucose and collagen.

Which difference explains the fact that  $\beta$ -glucose is soluble in water, while collagen is insoluble in water?

|           | $\beta$ -glucose                | Collagen                                     |
|-----------|---------------------------------|----------------------------------------------|
| <b>A.</b> | does not contain hydrogen bonds | contain hydrogen bonds                       |
| <b>B.</b> | monomer                         | polymer                                      |
| <b>C.</b> | absence of R-groups             | presence of R-groups                         |
| <b>D.</b> | contains polar group            | contains outward-facing hydrophobic R-groups |

8. Which combination of bonds contributes to the structure of the following molecules?

|           | Amylase                        | Amylopectin                    | Glucagon                       | Glycogen                       |
|-----------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| <b>A.</b> | ionic bond                     | $\alpha$ (1,6) glycosidic bond | peptide bond                   | $\alpha$ (1,4) glycosidic bond |
| <b>B.</b> | phosphodiester bond            | $\alpha$ (1,4) glycosidic bond | hydrogen bond                  | $\alpha$ (1,6) glycosidic bond |
| <b>C.</b> | $\alpha$ (1,4) glycosidic bond | $\alpha$ (1,6) glycosidic bond | $\alpha$ (1,6) glycosidic bond | $\beta$ (1,4) glycosidic bond  |
| <b>D.</b> | covalent bond                  | $\beta$ (1,4) glycosidic bond  | hydrophobic interactions       | $\alpha$ (1,6) glycosidic bond |

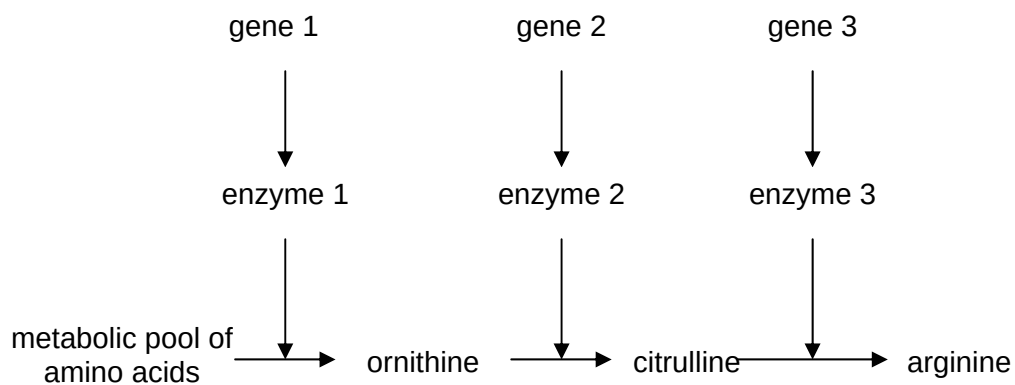
9. Beetroot cells contain a water-soluble red pigment. Two test-tubes were set up as described in the table.

|               |                                                                                    |
|---------------|------------------------------------------------------------------------------------|
| <i>tube 1</i> | pieces of washed raw beetroot in water                                             |
| <i>tube 2</i> | pieces of washed raw beetroot in water containing 3 drops of respiratory inhibitor |

After 30 minutes, the water in tube 2 contained red pigment but the water in tube 1 did not.

Which statement explains the colour of the water in tube 2?

- A. Pigment molecules passed out and were replaced by the inhibitor molecule.
  - B. The cell membranes were unable to retain the pigment.
  - C. The cell walls were made permeable to the pigment.
  - D. Water passed out of the cells by osmosis and carried the soluble pigment with it.
10. The diagram shows a metabolic pathway involving some amino acids.

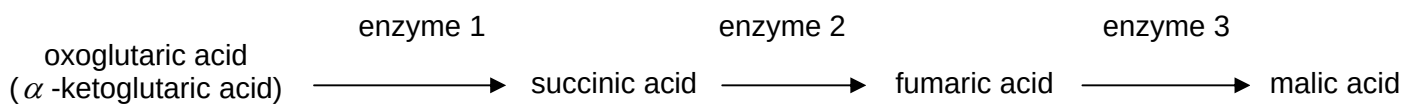


Three mutant strains of *Escherichia coli* each fail to produce a different enzyme in the pathway.

What should be supplied to the culture medium to enable growth of all three mutant strains?

- A. arginine
- B. citrulline
- C. citrulline and arginine
- D. citrulline and ornithine

11. The four acids shown below form part of an enzyme-catalysed sequence.



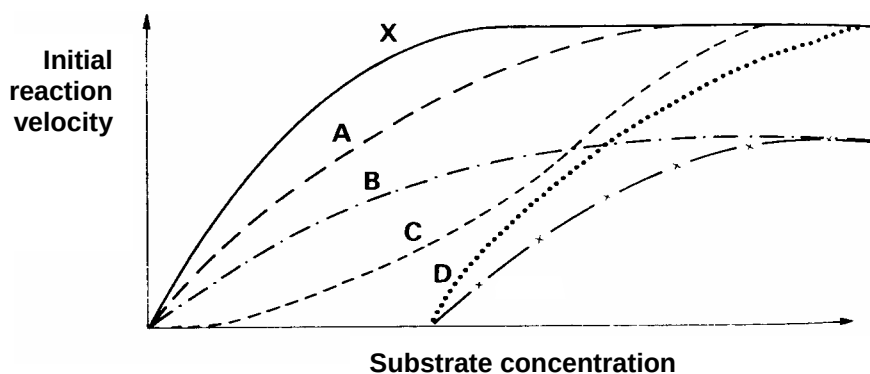
The addition of malonic acid results in no change in the concentration of oxoglutaric acid, an accumulation of succinic acid, and a near absence of both fumaric acid and malic acid.

Further addition of fumaric acid results in the formation of malic acid.

What does this information indicate about malonic acid?

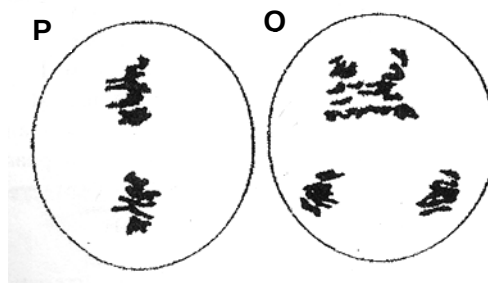
- A. It is an inhibitor of enzyme 1.
- B. It is an inhibitor of enzyme 2.**
- C. It catalyses the formation of succinic acid.
- D. It inhibits enzyme 3.

12. In the graph below, X represents the relationship between an enzyme and the concentration of its substrate under optimal conditions and without an inhibitor.



Which of the curves, A, B, C or D represents the expected result when the same experiment is conducted in the presence of a fixed, low concentration of non-competitive inhibitor?

13. The diagrams P and Q below show two stages in meiosis. (The chromosome outlines have been traced from photomicrographs.)



What are these stages?

|           | P                   | Q                  |
|-----------|---------------------|--------------------|
| A.        | metaphase I         | telophase I        |
| B.        | metaphase I         | telophase II       |
| <b>C.</b> | <b>metaphase II</b> | <b>anaphase II</b> |
| D.        | prophase II         | telophase I        |

14. If one ignores the effect of crossing over, then the possible number of different haploid cells which could arise by the meiotic division of a diploid cell containing 10 chromosomes (i.e.  $n=5$ ) is:

A. 2  
B. 4  
C. 8  
D. 32

15. Adenine comprises 0.32 of the nitrogenous bases in the DNA of cells from a bacterial clone.

What is the proportion of guanine in the DNA?

A. 0.12  
B. 0.18  
C. 0.32  
D. 0.36

16. An experiment is carried out to show that DNA replication is semi-conservative.

*Escherichia coli* cells whose DNA contains only 'heavy' nitrogen ( $^{15}\text{N}$ ) are grown in a culture medium containing the normal isotope of nitrogen ( $^{14}\text{N}$ ) and allowed to divide three times.

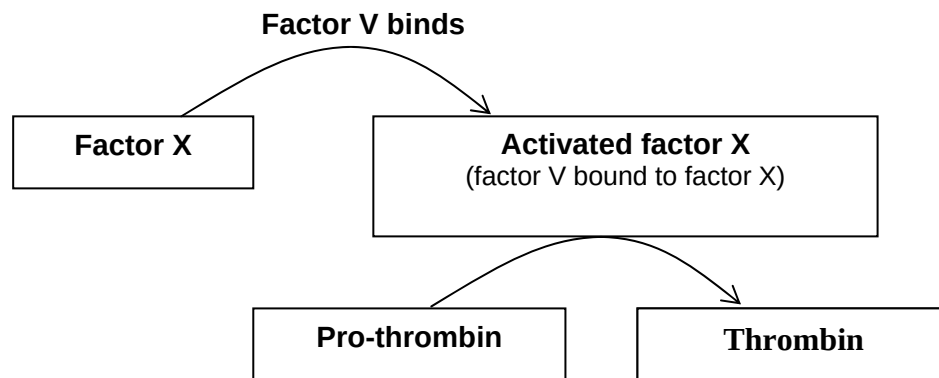
What will be the percentage of DNA molecules containing both forms of nitrogen after the third division?

A. 12.5%  
B. 25%  
C. 50%  
D. 100%

17. The basis for development of leukemia in some children treated with a retrovirus-derived gene therapy vector is:

A. Insertion of the viral genome adjacent to an oncogene  
B. Use of a virus that contains an oncogene  
C. Insertion causing a mutation of a tumor suppressor gene  
D. Stimulation of immune response to cell surface antigens introduced by an inserted gene

18. Multiple proteins are involved in the process of blood clotting. The diagram below shows how factor V functions as a co-factor in blood clotting.



Thrombin goes on to cleave fibrinogen to form fibrin, which helps in blood clot formation. Another protein that is involved in blood clotting is activated protein C (aPC), a natural anticoagulant that acts to limit the extent of clotting by cleaving and degrading factor V.

A mutation in factor V protein results in a variant protein called factor V Leiden. The Leiden variant of the protein is resistant to degradation by aPC.

Choose the option that shows the mode of inheritance of the Leiden allele, the correct type of mutation this is and the result this mutation will have on blood clotting.

|    | Mode of inheritance | Type of mutation | Result on blood clotting |
|----|---------------------|------------------|--------------------------|
| A. | Dominant            | Gain-of-function | Blood clots more easily  |
| B. | Dominant            | Loss-of-function | Blood clots less easily  |
| C. | Recessive           | Gain-of-function | Blood clots less easily  |
| D. | Recessive           | Loss-of-function | Blood clots more easily  |

19. How do viruses increase the occurrence of cancer?

- A. Viral DNA may be tumour suppressor genes that are integrated into the host cell genome and cause uncontrolled cell division.
- B. Viral DNA may disrupt oncogenes when integrated into the host cell genome and cause uncontrolled cell division.
- C. Viral DNA may be proto-oncogenes that cause cancer when integrated into the host cell genome.
- D. Viral DNA may be oncogenes that cause cancer when integrated into the host cell genome.

20. A group of scientists uses modified viruses as vectors for the introduction of foreign genes into bacteria.

In one experiment, a bacteriophage was put together with the protein coat of T4 phage and the DNA of a lambda phage. If this composite phage was allowed to infect bacteria, the phages produced in the bacteria will have

- A. the protein of lambda phage and the DNA of T4.
- B. the protein of T4 and the DNA of lambda phage.
- C. the protein and DNA of lambda phage.
- D. the protein and DNA of T4.

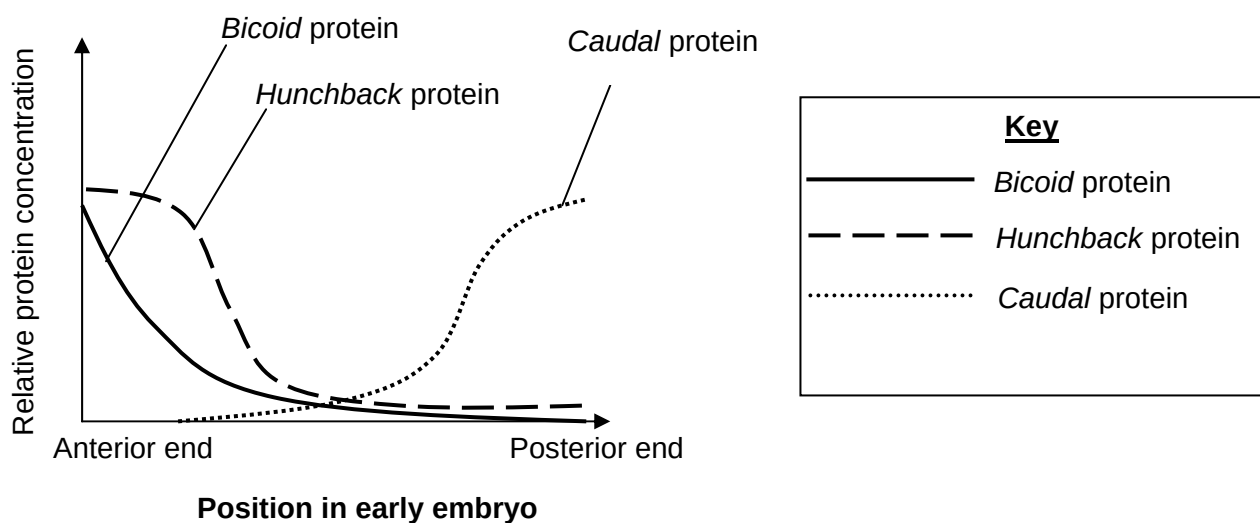


21. Which of the following is **incorrectly** matched with its function?

- A. Promoter – where RNA polymerase first binds to DNA
- B. Regulatory gene – binds to the repressor protein**
- C. Structural gene – synthesizes mRNA by transcription
- D. Operator – if unbound, allows RNA polymerase to bind to DNA

22. In *Drosophila*, three of the genes involved in embryological development are named *bicoid*, *caudal* and *hunchback*. *Bicoid* was thought to affect the expression of *hunchback* and *caudal*.

To determine the effect of *bicoid* on the other two genes, the levels of *bicoid*, *caudal* and *hunchback* proteins in a developing *Drosophila* embryo were determined. The results are shown in the diagram below.



Which of the following options best describes the possible mechanisms by which the ***bicoid* protein** might affect the expression of the *hunchback* and *caudal* genes?

|           | Effect on <i>hunchback</i>                                                                                       | Effect on <i>caudal</i>                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| A.        | <i>Bicoid</i> protein binds to <i>hunchback</i> enhancer and increases transcription of <i>hunchback</i> .       | <i>Bicoid</i> protein binds to <i>caudal</i> enhancer and increases transcription of <i>caudal</i> .     |
| B.        | <i>Bicoid</i> protein binds to <i>hunchback</i> silencer and decreases transcription of <i>hunchback</i> .       | <i>Bicoid</i> protein binds to <i>caudal</i> enhancer and increases transcription of <i>caudal</i> .     |
| <b>C.</b> | <b><i>Bicoid</i> protein binds to <i>hunchback</i> enhancer and increases transcription of <i>hunchback</i>.</b> | <b><i>Bicoid</i> protein binds to <i>caudal</i> mRNA and prevents translation of <i>caudal</i> mRNA.</b> |
| D.        | <i>Bicoid</i> protein binds to <i>hunchback</i> mRNA and prevents translation of <i>hunchback</i> mRNA.          | <i>Bicoid</i> protein binds to <i>caudal</i> silencer and decreases transcription of <i>caudal</i> .     |

23. A strain of toad has only one nucleolus in the nucleus of each cell instead of the usual two. When toads with one nucleolus per cell are mated, approximately a quarter of the offspring have two nucleoli per nucleus, half have one nucleolus per nucleus and a quarter have no nucleoli.

What is the most likely explanation of these results?

- A. The possession of one nucleolus is due to autosomal linkage.
- B. The possession of one nucleolus is due to the heterozygous condition.
- C. The allele for the presence of two nucleoli is recessive.
- D. The allele for the presence of two nucleoli is dominant.

24. In fruit flies, body colour and eye colour are governed by different gene loci. A cross between yellow-bodied white-eyed females and grey-bodied red-eyed males produced females which were grey-bodied and red-eyed. When the  $F_1$  females were backcrossed with their male parents, the resulting  $F_2$  males comprised the following:

1330 yellow-bodied white-eyed flies,  
 26 yellow-bodied red-eyed flies,  
 35 grey-bodied white-eyed flies, and  
 1297 grey-bodied red-eyed flies

Which of the following explains the above results?

- A. In fruit flies, the male is the homogametic sex.
- B. The two gene loci segregate by independent assortment.
- C. The two gene loci are sex-linked and crossing-over had taken place during gamete formation.
- D. The allele for grey body is dominant over the allele for yellow body and the allele for white eyes is dominant over the allele for red eyes.

25. Which of the following statements are true?

- 1 Continuous variation is controlled by the additive effects of polygenes.
- 2 Continuous variation is always affected by the environment.
- 3 Discontinuous variation is sometimes affected by the environment.
- 4 Discontinuous variation exhibits a normal distribution curve.

- A. 1 and 2
- B. 1 and 3
- C. 3 and 4
- D. None of the above

26. Labrador Retrievers, a popular breed of dog, can have several possible coat colours – black, chocolate or yellow.

Individuals with the genotype BBEE are black, while individuals with the genotype bbee are yellow. A cross between individuals with genotypes BBEE and bbee resulted in offspring that were all black in colour.

The F<sub>1</sub> offspring were sibling mated, and the coat colours of the resulting F<sub>2</sub> offspring were recorded.

| Coat colour    | No. |
|----------------|-----|
| Black coat     | 565 |
| Chocolate coat | 184 |
| Yellow coat    | 254 |

Which of the following options correctly describes the mode of inheritance of coat colour in Labrador Retrievers?

- A. Collaboration
- B. Co-dominance
- C. Dominant epistasis
- D. Recessive epistasis**

27. Six tubes were set up as shown in the table.

| tube | Contents                                |
|------|-----------------------------------------|
| 1    | Glucose + homogenized plant cells       |
| 2    | Glucose + mitochondria                  |
| 3    | Glucose + cytoplasm lacking organelles  |
| 4    | Pyruvate + homogenized animal cells     |
| 5    | Pyruvate + mitochondria                 |
| 6    | Pyruvate + cytoplasm lacking organelles |

After incubation, each sample was analysed to determine the presence of carbon dioxide and ethanol.

In which tubes are lactate most likely to be present?

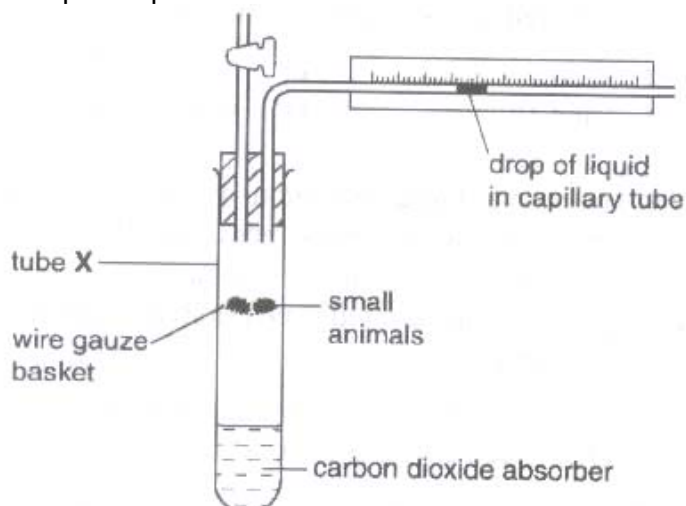
- A. 1 and 3 only
- B. 2, 3, 5 and 6 only
- C. 4, 5, and 6 only
- D. 3 and 6 only**

28. A particular type of autoimmune disease causes antibodies to destroy acetylcholine receptors of neurons.

What effect will this have on the functioning of the nervous system?

- A. Presynaptic neurons will be unable to release neurotransmitters into the synaptic clefts.
- B. Neurons will be unable to propagate action potentials along their axons.
- C. Depolarized neurons will be unable to reestablish an ionic gradient across their membranes.
- D. Postsynaptic neurons will be unable to detect signals from presynaptic neurons.**

29. The diagram shows a simple respirometer.



The changes in gas volume in the tube are measured at intervals.

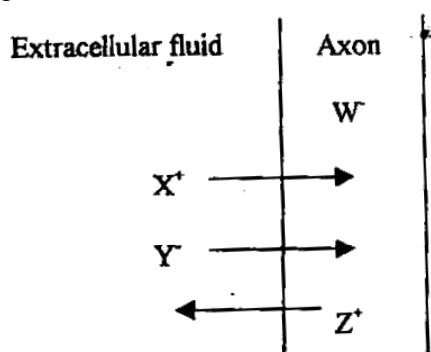
| time<br>(minutes) | gas volume with<br>carbon dioxide<br>absorber ( $\text{cm}^3$ ) | gas volume without<br>carbon dioxide<br>absorber ( $\text{cm}^3$ ) |
|-------------------|-----------------------------------------------------------------|--------------------------------------------------------------------|
| 0                 | 0.0                                                             | 0.0                                                                |
| 10                | -0.4                                                            | -0.1                                                               |
| 20                | -0.8                                                            | -0.2                                                               |
| 30                | -1.2                                                            | -0.3                                                               |

Tube X contains 2 g of small animals.

What is the carbon dioxide output per g per hour for these organisms?

- A.  $0.9 \text{ cm}^3$
- B.  $1.8 \text{ cm}^3$
- C.  $2.4 \text{ cm}^3$
- D.  $4.8 \text{ cm}^3$

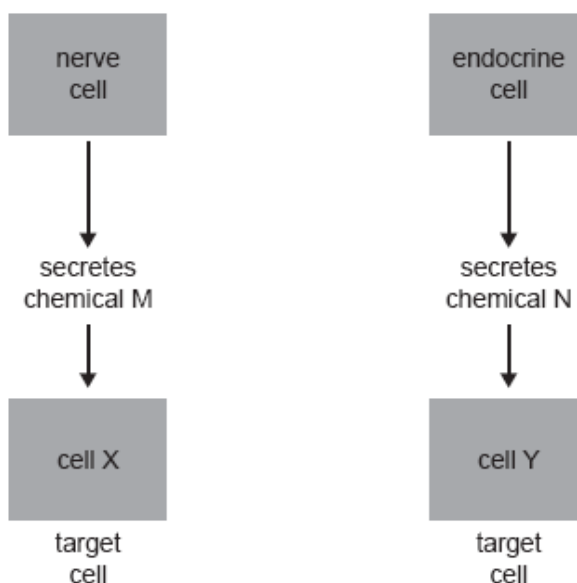
30. The diagram shows ionic movements during an action potential



Which one of the options (A-D) correctly identifies the ions  $W^-$ ,  $X^+$ ,  $Y^-$  and  $Z^+$ ?

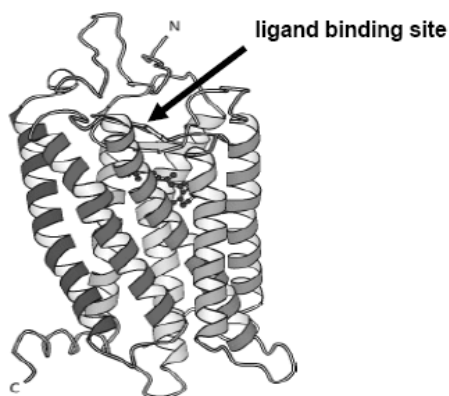
|    | $W^-$              | $X^+$     | $Y^-$              | $Z^+$     |
|----|--------------------|-----------|--------------------|-----------|
| A. | Organic acid anion | Chloride  | Sodium             | Potassium |
| B. | Organic acid anion | Potassium | Chloride           | Sodium    |
| C. | Organic acid anion | Sodium    | Chloride           | Potassium |
| D. | Chloride           | Potassium | Organic acid anion | Sodium    |

31. The diagram shows how a nerve cell and an endocrine cell interact with cell X and cell Y respectively.



Which of the following is true of both interactions?

- A. When both cells X and Y are stimulated, phosphorylation cascades must be initiated.
  - B. Chemical M is secreted faster than chemical N.
  - C. Diffusion is involved in the movement of chemicals M and N to their respective target cells.
  - D. Both chemicals M and N must bind to cell-surface receptors on cell X and Y respectively.
32. The figure below shows a common cell membrane receptor, which comprises of a single polypeptide chain.



Which statement correctly describes how the receptor responds upon ligand binding?

- A. Aggregates and forms a dimer
- B. Undergoes conformation change and binds to G-protein
- C. Undergoes conformation change and allows ions to flow through
- D. Enters the nucleus and binds to DNA

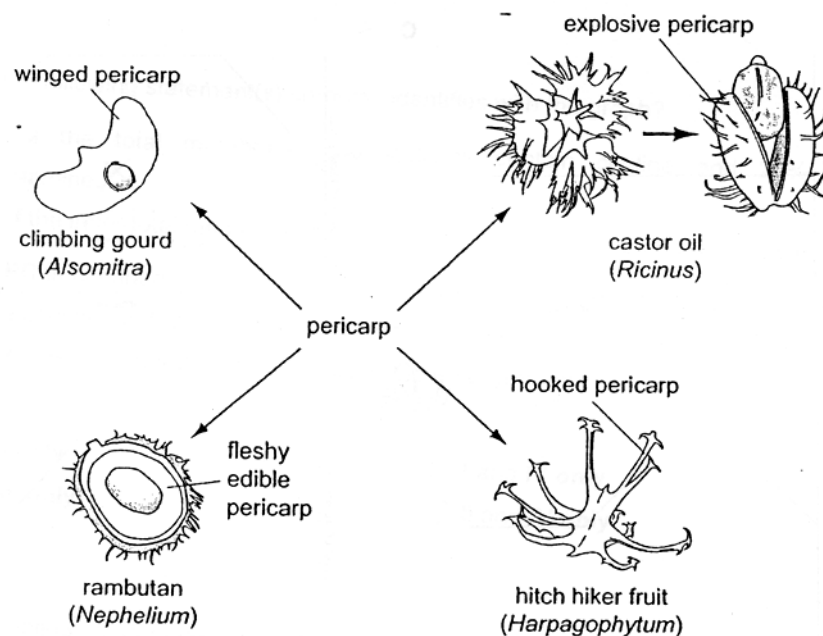
33. Which of the following is **not** a possible mode of action of antidiuretic hormones, synthesized by the rough endoplasmic reticulum of the hypothalamus, on target cells?

- A. Binding to G-protein coupled receptors on target cell membrane
- B. Binding to intracellular receptors in target cells
- C. Initiating phosphorylation cascade in target cells
- D. Initiating conversion of ATP to cAMP

34. Which of the following has evolved mainly as a result of artificial selection?

- A. Darker colouring of the peppered moth near industrial areas
- B. Increased production of antibiotics by the fungus *Penicillium* species
- C. Increased resistance of houseflies to the insecticide DDT
- D. Increased tolerance of lichens to heavy metals on the bark around mine workings

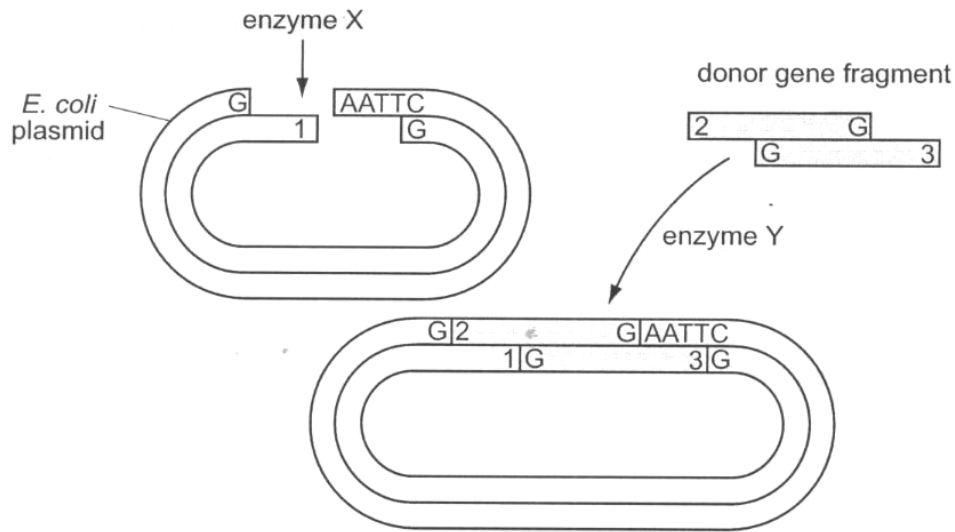
35. The diagram illustrates variation in the pericarp (fruit wall) for a variety of methods of seed dispersal.



What do these examples illustrate?

- A. The adaptive radiation of analogous structures showing divergent evolution.
- B. The adaptive radiation of analogous structures showing convergent evolution.
- C. The adaptive radiation of homologous structures showing divergent evolution.
- D. The adaptive radiation of homologous structures showing convergent evolution.

36. The diagram shows some stages in making recombinant DNA using *E. coli* plasmids.



Which are the correct enzymes X and Y and order of the bases at 1, 2 and 3?

|    | X                  | Y                  | 1    | 2    | 3    |
|----|--------------------|--------------------|------|------|------|
| A. | Ligase             | Restriction enzyme | TTAA | AATT | TTAA |
| B. | Restriction enzyme | Ligase             | TTAA | AATT | TTAA |
| C. | Ligase             | Restriction enzyme | AATT | TTAA | AATT |
| D. | Restriction enzyme | Ligase             | AATT | TTAA | AATT |

37. What are the normal functions of stem cells in a living human?

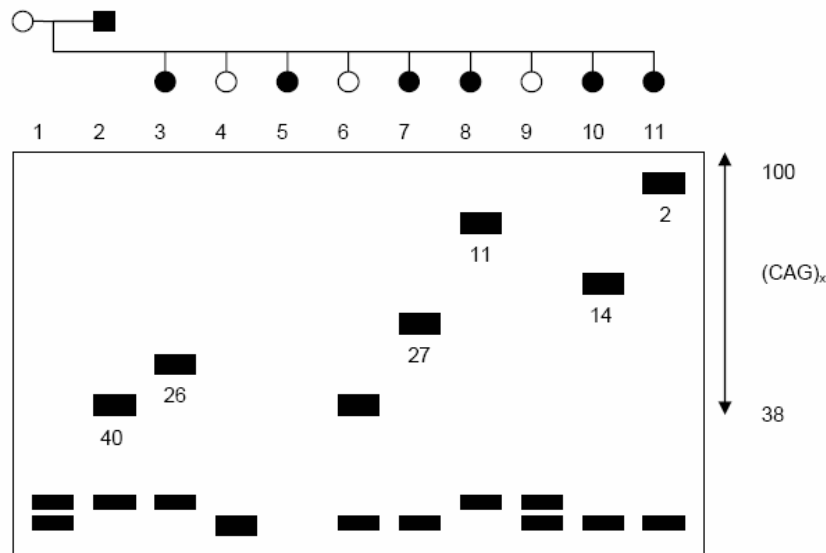
|    | Functions                                                                   |                                                                     |                                                                     |
|----|-----------------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------|
| A. | Differentiating into many kinds of cells in a 3 – 5 day old human embryo    | Producing insulin in pancreas damaged by type 1 diabetes            | Cells that can differentiate into bone cells in the skeletons       |
| B. | Differentiating into many kinds of cells in a 3 – 5 day old human embryo    | Producing red blood cells worn out by normal wear and tear          | Cells that can differentiate into cardiac muscle cells in the heart |
| C. | Differentiating into only one kind of cells in a 3 – 5 day old human embryo | Producing dopamine in the brains of people with Parkinson's         | Cells that can differentiate into cartilage cells in the joints     |
| D. | Differentiating into only one kind of cells in a 3 – 5 day old human embryo | Producing differentiated cells that can be used to screen new drugs | Cells that can differentiate into nerve cells in the brain          |

38. Which of the following disorders would gene therapy be **least** effective?

- A. Cystic fibrosis
- B. Huntington's disease
- C. Sickle cell anaemia
- D. Type II diabetes

39. The data below shows the results of electrophoresis of PCR fragments amplified using primers for the site which has been shown to be altered in Huntington's disease. The inherited mutation in the Huntington's disease gene abnormally repeats the nucleotide sequence CAG from 36 to more than 120 times.

The male parent, as shown by the black box, got Huntington's disease when he was 40 years old. Six of his children (individuals 3, 5, 7, 8, 10, 11) suffer from Huntington's disease, and the age at which the symptoms first began is shown by the number below the band from the PCR fragment.



What is the likely outcome for the normal individuals 4, 6, and 9?

- A. Individuals 4 and 9 do not have the trait, and will not get Huntington's disease, but individual 6 is likely to start the disease when he reaches his father's age of 40.
- B. Individuals 4, 6, and 9 have not inherited the defect causing Huntington's disease.
- C. Individuals 4, 6, and 9 will still develop Huntington's disease at some point in their lives, since the disease is inherited as a dominant trait.
- D. Two of the three will develop the disease, since it is inherited as a dominant trait, but the data does not allow you to predict which two.

40. Which of the following are reasons for scientists to employ the method of plant cloning?

- 1 change the phenotype of an organism
- 2 create smaller genetic changes at a much more rapid pace
- 3 production of large amount of pharmaceuticals
- 4 selective breeding is too slow

- A. 1 and 2
- B. 1 and 3
- C. 1, 2 and 3
- D. 1, 2, 3 and 4